Course: Cloud Computing

Course Code: CSA1514

Slot: B

Faculty: **Govindasamy C**

Submitted By

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Short Answer Test

1. Define cloud computing and explain any four characteristics.

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet instead of maintaining them on a local computer or organization's own infrastructure.

Four characteristics:

1. **On-demand self-service:** Users can obtain computing resources whenever required without direct human interaction with the service provider.
2. **Broad network access:** Cloud services can be accessed through the internet using laptops, mobiles, tablets, and other devices.
3. **Resource pooling:** Computing resources such as storage, memory, and processing power are shared among multiple users efficiently.
4. **Rapid elasticity:** Resources can be quickly increased or decreased depending on demand.

Difference from on-premises: In traditional on-premises computing, an organization purchases and maintains its own hardware and software. In cloud computing, resources are obtained as services from a cloud provider.

2. Trace the evolution of cloud computing.

Cloud computing developed through several stages:

1. Hardware evolution:
Early computers were large and expensive mainframes. Improvements in processors, storage, networking, and server technology made computing more powerful and affordable.

2. Virtualization:
Virtualization allowed one physical server to run multiple **virtual machines (VMs)**. This improved hardware utilization and reduced costs.

3. Internet evolution:
Faster and more reliable internet connections made it possible to access computing resources from remote locations.

4. Web and software evolution:
Web applications and distributed computing allowed software and services to be delivered through browsers instead of being installed only on local computers.

5. Modern cloud computing:
These developments enabled modern cloud platforms to provide scalable servers, storage, databases, applications, AI services, and other resources over the internet.

3. Explain the role of server virtualization in cloud computing.

Server virtualization is the process of dividing one physical server into multiple independent **virtual machines (VMs)** using software called a **hypervisor**.

For example:

```
Physical Server
     ↓
  Hypervisor
     ↓
-------------------
VM 1   VM 2   VM 3
```

Linux Windows Linux

It is important in cloud computing because it provides:

- Better utilization of physical hardware
- Isolation between users
- Easy creation and removal of virtual servers
- Scalability
- Reduced hardware costs

Virtualization vs Cloud Computing

Virtualization:

1. Virtualization is a **technology**.
2. It creates **virtual resources** from physical hardware.
3. It uses **virtual machines (VMs) and hypervisors**.
4. It helps in **efficient utilization of hardware resources**.
5. It can work **without cloud computing**.
6. Example: **VMware, VirtualBox**.

Cloud Computing:

1. Cloud computing is a **service model**.
2. It delivers **computing resources over the internet/network**.
3. It can use **virtualization, containers, and automation**.
4. It provides **on-demand and scalable resources**.
5. Users can access cloud services **from anywhere through the internet**.
6. Example: **AWS, Microsoft Azure, Google Cloud**

Therefore, **virtualization is an enabling technology**, while **cloud computing is a model for delivering computing services**.

4. Discuss the role of data centers in delivering cloud services.

A **data center** is a physical facility containing servers, storage systems, networking equipment, power systems, and cooling systems.

Data centers form the physical foundation of cloud computing. They:

- Store and process users' data.
- Host cloud applications and virtual machines.

- Provide large amounts of computing power.
- Provide high-speed network connectivity.
- Use backup power and redundant systems to improve availability.
- Use cooling systems to prevent hardware from overheating.
- Implement physical and digital security measures.

Cloud providers operate large data centers in different geographical regions so that services can be delivered reliably and at scale.

5. Differentiate IaaS, PaaS, SaaS, and XaaS

1. IaaS – Infrastructure as a Service

- Provides **virtual servers, storage, and networking**.
- Users manage the operating system and applications.
- Provides more control over computing infrastructure.
- **Example:** Amazon EC2.

2. PaaS – Platform as a Service

- Provides a **platform for developing and deploying applications**.
- Developers don't need to manage the underlying hardware.
- Provides development tools, runtime environments, and databases.
- **Example:** Google App Engine.

3. SaaS – Software as a Service

- Provides **ready-to-use software over the internet**.
- Users don't need to install or maintain the software.
- Usually accessed through a web browser or application.
- **Example:** Google Docs.

4. XaaS – Anything/Everything as a Service

- Refers to **any IT service delivered through the cloud**.
- It includes IaaS, PaaS, SaaS and many other cloud services.
- Examples include Database as a Service, Security as a Service, and Backup as a Service.
- **Example:** Backup as a Service (BaaS).